Compactification of Separated Morphisms of Finite Type

A lecture on Nagata's compactification theorem

Abstract

This lecture presents a complete route to Nagata's compactification theorem: *a separated, quasi-compact morphism locally of finite type over a quasi-compact quasi-separated base factors as an open immersion followed by a proper morphism.* The material is organised in dependency order. We first develop the formal calculus of compactifications of morphisms and its three stability constructions (immersions, closed subschemes, base change). We then treat the base case, in which the source is quasi-affine, by embedding it into a relative affine space and closing it up inside a relative projective space. The bulk of the lecture is the geometry needed to glue two compactifications: the theory of finite-type quasi-coherent ideal sheaves and their extension across quasi-compact opens; admissible blow-up centres, their composition, extension, and density properties; the identification of the strict transform with the blow-up of a pulled-back centre; elimination of indeterminacy for partial sections, proved along the route of Deligne and Conrad through Zariski's Main Theorem; the resulting modification theorems (common modifications, graph extensions, isomorphism near a boundary closure); and finally the two-open compactification theorem. Induction over a finite affine open cover with dense total intersection gives the theorem over a Noetherian base, absolute Noetherian approximation removes the Noetherian hypothesis for morphisms of finite presentation, and the separated finite-presentation envelope of a morphism of finite type completes the proof. A closing section delimits precisely what the surveyed material establishes and which results it takes from outside; a summary table and a dependency mind map conclude the lecture.

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1 Conventions and notation

All schemes and morphisms are schemes and morphisms of schemes. We write Spec for the spectrum of a commutative ring, and we freely use the categorical fibre product $X \times_S Y$ of schemes.

A scheme is *quasi-compact* if its underlying topological space is, and *quasi-separated* if the intersection of two quasi-compact opens is quasi-compact; *quasi-compact quasi-separated* abbreviates the conjunction of the two. A morphism $f\colon X \to Y$ is *quasi-compact* if preimages of quasi-compact opens are quasi-compact, *separated* if its diagonal is a closed immersion, *locally of finite type* (resp. *locally of finite presentation*) if it is so on affine charts, and *proper* if it is separated, of finite type and universally closed. A morphism is *quasi-finite* in the sense used below if it is locally quasi-finite, that is, has discrete fibres.

We write X_{op} for the lattice of open subschemes of X; for an open $U \subseteq X$ we write $\iota_U\colon U \to X$ for the inclusion, and $f^{-1}U$ for the preimage open. For $f\colon X \to Y$ and an open $V \subseteq Y$, $f|_V\colon f^{-1}V \to V$ denotes the restriction.

Ideal sheaves. A *quasi-coherent ideal sheaf* $\mathcal{I}$ on a scheme X is recorded by its ideals of sections $\mathcal{I}(U) \subseteq \Gamma(X,U)$ over the affine opens U of X; these ideals form a complete lattice, with $\bot$ the zero ideal sheaf and $\top$ the unit ideal sheaf, and with products $\mathcal{I}\cdot\mathcal{L}$ and suprema $\mathcal{I}\sqcup\mathcal{L}$ computed sectionwise. We write $V(\mathcal{I})$ for the support of $\mathcal{I}$, a closed subset of X, and $V(\mathcal{I})^c$ for its open complement. For a morphism $f\colon Y \to X$ we write $\mathcal{I}\cdot\mathcal{O}_Y$ for the inverse image ideal sheaf of $\mathcal{I}$ on Y; it satisfies $(\mathcal{I}\cdot\mathcal{L})\cdot\mathcal{O}_Y = (\mathcal{I}\cdot\mathcal{O}_Y)\cdot(\mathcal{L}\cdot\mathcal{O}_Y)$ and $V(\mathcal{I}\cdot\mathcal{O}_Y) = f^{-1}V(\mathcal{I})$. The closed subscheme cut out by $\mathcal{I}$ is denoted $V(\mathcal{I})$ as well when no confusion arises, with closed immersion $V(\mathcal{I}) \to X$. An ideal sheaf is *invertible* if it is locally generated by a single nonzerodivisor.

Scheme-theoretic images. For a quasi-compact morphism $h\colon V \to X$ we write $\ker h$ for the quasi-coherent ideal sheaf of sections vanishing on V, $\overline{h(V)}$ (or $\operatorname{im} h$) for the closed subscheme it cuts out, $h_{\to}\colon V \to \operatorname{im} h$ for the induced morphism and $\iota_{\operatorname{im} h}\colon \operatorname{im} h \to X$ for the closed immersion. A morphism h is *scheme-theoretically dominant* if $\ker h = \bot$, i.e. if $\operatorname{im} h = X$; we also say that an open $U \subseteq X$ is *scheme-theoretically dense* if ι_U is scheme-theoretically dominant.

Blow-ups. For a quasi-coherent ideal sheaf $\mathcal{I}$ on X we write $\mathrm{Bl}_{\mathcal{I}}(X)$ for the blow-up of X in $\mathcal{I}$ and $\pi_{\mathcal{I}}\colon \mathrm{Bl}_{\mathcal{I}}(X) \to X$ for its structure morphism.

Relative affine and projective spaces. For an index type σ and a scheme S we write $\mathbb{A}(\sigma;S)$ for the relative affine space over S with coordinates indexed by σ, and $\mathbb{A}(\sigma;S) \to S$ for its structure morphism; $\mathbb{P}(\sigma;S)$ is introduced in §3.

Statements are numbered by section, and every named result carries a label so that it may be referred to from the summary table of §14 and from the dependency diagrams of §15. A displayed block set between two horizontal rules and headed **Auxiliary mathematical input used here** records, at the point where it is first used, a mathematical fact which the lecture takes as given rather than proves.

2 Compactifications of morphisms

Definition 2.1 (Compactification)**.** Let $f\colon X \to Y$ be a morphism. A *compactification* of f is a triple (Z,g,i) consisting of a scheme Z, a proper morphism $g\colon Z \to Y$ and an open immersion $i\colon X \to Z$ such that

$$f = g \circ i.$$

We say that f *admits a compactification*, or is *compactifiable*, if such a triple exists.

The first proposition collects what a compactification forces on f itself and on its open immersion.

Proposition 2.2 (Properties of compactifiable morphisms)**.** *Let (Z, g, i) be a compactification of $f\colon X \to Y$. Then:*

(i) *f is separated;*

(ii) *f is locally of finite type;*

(iii) *i is quasi-compact if and only if f is quasi-compact;*

(iv) *if f is proper, then i is proper, hence a closed immersion, and its range is open and closed in Z.*

Proof. Items (i) and (ii) hold because g is proper, hence separated and locally of finite type, i is an open immersion, hence separated and locally of finite type, and both classes are stable under composition. For (iii), if i is quasi-compact then so is $g \circ i = f$; conversely if $g \circ i$ is quasi-compact then i is, because g is separated. Item (iv) is the same cancellation for properness: $g \circ i$ proper and g separated force i proper; a proper open immersion is a closed immersion, and a morphism which is both an open and a closed immersion has open and closed range. □

Proposition 2.3 (Elementary constructions)**.** (i) *A proper morphism f is a compactification of itself, via (X, f, id_X).*

(ii) *A compactification of f is a compactification of any morphism equal to f.*

(iii) (Restriction to open subschemes.) *If (Z, g, i) compactifies $f\colon X \to Y$ and $e\colon X' \to X$ is an open immersion, then $(Z, g, i \circ e)$ compactifies $f \circ e$. In particular compactifiability is inherited by open subschemes of the source.*

2.1 Compactification by closing up an immersion

Auxiliary mathematical input used here. For a quasi-compact morphism $h\colon V \to X$ the scheme-theoretic image $\operatorname{im} h$ exists as a closed subscheme of X, with a canonical factorization $h = \iota_{\operatorname{im} h} \circ h_{\to}$. If h is moreover an immersion, then $h_{\to}\colon V \to \operatorname{im} h$ is an open immersion.

Construction 2.4 (Compactification from an immersion)**.** Let $g\colon X \to Y$ be a quasi-compact immersion and let $p\colon Y \to S$ be proper. The factorization of $p \circ g$ through the scheme-theoretic image of g,

$$X \xrightarrow{g_{\to}} \operatorname{im} g \xrightarrow{\iota_{\operatorname{im} g}} Y \xrightarrow{p} S,$$

is a compactification of $p \circ g$: the first morphism is an open immersion because g is a quasi-compact immersion, and $p \circ \iota_{\operatorname{im} g}$ is proper as the composite of a closed immersion with a proper morphism.

Theorem 2.5 (Immersion into a proper scheme)**.** *Let $u\colon X \to S$ be quasi-compact and suppose that $u = p \circ g$ for an immersion $g\colon X \to Y$ and a proper morphism $p\colon Y \to S$. Then u admits a compactification.*

Proof. Since $u = p \circ g$ is quasi-compact and p is separated, g is quasi-compact, so Construction 2.4 applies. □

Corollary 2.6 (Compactifiability passes to closed subschemes)**.** *Let $i\colon X \to X'$ be a closed immersion and let $f'\colon X' \to S$ be quasi-compact and compactifiable. Then $f' \circ i$ admits a compactification; one may take for its ambient scheme the scheme-theoretic image of X inside the ambient scheme of a compactification of f'.*

Proof. Let (Z, g, j) compactify f'. By Proposition 2.2(iii) the open immersion j is quasi-compact, hence $j \circ i$ is a quasi-compact immersion into the scheme Z, which is proper over S. Now apply Theorem 2.5, using $f' \circ i = g \circ (j \circ i)$. □

2.2 Base change

Construction 2.7 (Base change of a compactification)**.** Let

$$\begin{array}{ccc} X & \xrightarrow{h} & Y \\ \downarrow u & & \downarrow p \\ S & \xrightarrow{t} & T \end{array}$$

be a cartesian square and let (Z, g, i) be a compactification of p. Then $Z \times_T S \to S$ is proper, the canonical morphism $X \to Z \times_T S$ determined by $h \circ i$ and u is an open immersion, and

$$X \longrightarrow Z \times_T S \longrightarrow S$$

is a compactification of u.

Lemma 2.8. *In the situation of Construction 2.7, the square formed by h, the morphism $X \to Z \times_T S$, the open immersion i and the first projection $Z \times_T S \to Z$ is cartesian. Consequently $X \to Z \times_T S$ is an open immersion, open immersions being stable under base change.*

Theorem 2.9 (Compactifiability is stable under base change)**.** *If $p\colon Y \to T$ admits a compactification and $u\colon X \to S$ is a base change of p along a morphism $t\colon S \to T$, then u admits a compactification.*

3 Relative projective space and the quasi-affine base case

3.1 Relative projective space

Let σ be an index type. The polynomial ring $\mathbb{Z}[T_i : i \in \sigma \sqcup \{*\}]$ carries its total-degree grading.

Definition 3.1 (Relative projective space)**.** The *integral model* is $\mathbb{P}^\sigma_{\mathbb{Z}} = \operatorname{Proj} \mathbb{Z}[T_i]$ for the total-degree grading. For a scheme S, the *relative projective space* is the base change

$$\mathbb{P}(\sigma; S) \;=\; \mathbb{P}^\sigma_{\mathbb{Z}} \times_{\operatorname{Spec} \mathbb{Z}} S,$$

with its structure morphism $\mathbb{P}(\sigma; S) \to S$.

Theorem 3.2 (Properness of relative projective space)**.** *For finite σ the structure morphism $\mathbb{P}(\sigma; S) \to S$ is proper.*

Proof. The morphism $\mathbb{P}^\sigma_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$ is proper, and $\mathbb{P}(\sigma; S) \to S$ is one of its base changes; properness is stable under base change. □

Lemma 3.3 (Dehomogenization at a variable)**.** *Let R be a commutative ring and let $R[T_*, T_i : i \in \sigma]$ carry its total-degree grading. The degree-zero part of the localization away from the variable T_* is canonically isomorphic to the polynomial ring in the remaining variables:*

$$\left(R[T_*, T_i]_{T_*}\right)_0 \xrightarrow{\sim} R[T_i : i \in \sigma], \qquad \frac{a}{T_*^k} \longmapsto a(1, T_i),$$

for a homogeneous of degree k; the inverse sends T_i to T_i/T_*.

Lemma 3.4 (The standard chart)**.** *The morphism $\mathbb{A}(\sigma; S) \to \mathbb{P}(\sigma; S)$ induced on the integral model by $T_i \mapsto T_i/T_*$ is an open immersion, and it is a morphism over S.*

Proof. On the integral model the chart is the open immersion of the affine chart identified by Lemma 3.3. The relative statement is obtained by base change to S, the square exhibiting the relative chart as a base change of the integral one being cartesian; open immersions are stable under base change. □

3.2 Quasi-affine sources

Theorem 3.5 (Immersion of a quasi-affine scheme into affine space)**.** *Let X be a quasi-affine scheme and let $\pi\colon X \to S$ be locally of finite type. Then there are a finite index type σ and an immersion $i\colon X \to \mathbb{A}(\sigma; S)$ over S, i.e. with π equal to the composite of i with the structure morphism of $\mathbb{A}(\sigma; S)$. No hypothesis on S is needed.*

Sketch. At each point of X one chooses a basic affine open of X contained in the preimage of an affine open of S; over such a pair the morphism is described by finitely many sections. Quasi-affineness of X and finite type of π provide finitely many global sections of $\mathcal{O}_X$, containing a unit on the chart, which generate the sections over the chart as an algebra over the sections of the base. The resulting morphism to $\mathbb{A}(\sigma; S)$ is a closed immersion into the basic open of one of the coordinates, hence an immersion. □

Construction 3.6 (Compactification of a quasi-affine source)**.** Let $f\colon X \to S$ be quasi-compact and let $i\colon X \to \mathbb{A}(\sigma; S)$ be an immersion over S with σ finite. Composing i with the standard chart $\mathbb{A}(\sigma; S) \to \mathbb{P}(\sigma; S)$ of Lemma 3.4 exhibits f as an immersion into a scheme proper over S (Theorem 3.2); closing up its image inside $\mathbb{P}(\sigma; S)$ as in Construction 2.4 yields a compactification of f.

Theorem 3.7 (The quasi-affine case)**.** *Let $f\colon X \to S$ be quasi-compact and locally of finite type with X quasi-affine. Then f admits a compactification. In particular this holds when X is affine.*

Proof. Combine Theorem 3.5 with Construction 3.6. □

This is the base case of the induction of §10.

4 Quasi-coherent ideal sheaves of finite type

Definition 4.1 (Finite type)**.** Let $\mathcal{I}$ be a quasi-coherent ideal sheaf on X and let $O \subseteq X$ be an open. We say $\mathcal{I}$ is *of finite type on O* if $\mathcal{I}(U)$ is a finitely generated ideal for every affine open $U \subseteq O$, and *of finite type* if this holds for every affine open of X.

Proposition 4.2 (Stability)**.** *The zero and unit ideal sheaves are of finite type. Finite-typeness on an open is finite-typeness of the restriction to that open; it is local on the target, i.e. stable under finite suprema and under suprema of families of opens. On a locally Noetherian scheme every quasi-coherent ideal sheaf is of finite type. Finite-typeness is stable under inverse image along an arbitrary morphism.*

Theorem 4.3 (Extension of finite-type ideal sheaves)**.** *Let X be a quasi-compact scheme.*

(i) *Let $\mathcal{I}_0 \leq \mathcal{H}_0$ be quasi-coherent ideal sheaves on X, let $W \subseteq X$ be a quasi-compact open, and assume $\mathcal{I}_0$ is of finite type on W. Then there is a finite-type quasi-coherent ideal sheaf $\mathcal{I} \leq \mathcal{H}_0$ on X with $\mathcal{I}|_W = \mathcal{I}_0|_W$.*

(ii) *In particular, every finite-type quasi-coherent ideal sheaf $\mathcal{J}$ on a quasi-compact open $W \subseteq X$ extends to a finite-type quasi-coherent ideal sheaf $\mathcal{I}$ on X with $\mathcal{I}|_W = \mathcal{J}$.*

Sketch. One enlarges the open on which the ideal sheaf is of finite type one affine open at a time; a finite induction over an affine cover then concludes. The binary gluing step is order-theoretic: the glued ideal sheaf is the intersection of the maximal quasi-coherent extensions of the current ideal sheaf and of a finitely generated approximation on the new affine piece, intersected with the ambient ideal sheaf. Its restrictions are computed by flat base change of scheme-theoretic images, and the approximation on the new affine piece is obtained by clearing denominators on a finite cover of the overlap by basic opens. □

Theorem 4.4 (Extension along an affine morphism)**.** *Let X be quasi-compact, let $\mathcal{H}_0$ be a quasi-coherent ideal sheaf on X and let $j\colon Z \to X$ be an affine morphism such that $\mathcal{H}_0 \cdot \mathcal{O}_Z$ is of finite type. Then there is a finite-type $\mathcal{I} \leq \mathcal{H}_0$ on X with $\mathcal{I} \cdot \mathcal{O}_Z = \mathcal{H}_0 \cdot \mathcal{O}_Z$. If moreover $V \subseteq X$ is a quasi-compact open with $V \cap V(\mathcal{H}_0) = \emptyset$, then $\mathcal{I}$ may be chosen with $V \cap V(\mathcal{I}) = \emptyset$ as well.*

Sketch. Cover X by finitely many affine opens W; over each of them choose finitely many sections of $\mathcal{H}_0$ already generating its inverse image over the affine open $j^{-1}W$, extend the ideal sheaf they span to a finite-type ideal sheaf inside $\mathcal{H}_0$ by Theorem 4.3, and sum over the cover. □

Theorem 4.5 (Finite-type ideal sheaves are compact; directedness)**.** *Let X be a quasi-compact scheme.*

(i) *If $\mathcal{A}$ is of finite type and $\mathcal{A} \leq \bigvee_i \mathcal{B}_i$ for a directed family $(\mathcal{B}_i)_i$ of quasi-coherent ideal sheaves, then $\mathcal{A} \leq \mathcal{B}_i$ for some i.*

(ii) *Every quasi-coherent ideal sheaf $\mathcal{H}$ on a quasi-compact quasi-separated scheme is the supremum of the directed family of its finite-type quasi-coherent subsheaves; the same holds for the cofinal subfamily of those subsheaves which restrict to the unit ideal on a prescribed quasi-compact open.*

Theorem 4.6 (Boundary ideals)**.** *Let X be quasi-compact quasi-separated and let $U \subseteq X$ be a quasi-compact open. Then there is a finite-type quasi-coherent ideal sheaf $\mathcal{I}$ on X with $V(\mathcal{I}) = X \setminus U$. On a locally Noetherian scheme the same holds for an arbitrary open U, no quasi-compactness being needed.*

Sketch. The vanishing ideal of the reduced closed complement of U restricts to the unit ideal on U; Theorem 4.3 produces a finite-type subsheaf with the same restriction to U, and the support of such a subsheaf is exactly the complement of U. □

Lemma 4.7 (A power inside an ideal sheaf)**.** *Let X be quasi-compact, let $\mathcal{A}, \mathcal{B}$ be quasi-coherent ideal sheaves with $\mathcal{B}$ of finite type, and assume $V(\mathcal{A}) \subseteq V(\mathcal{B})$. Then $\mathcal{B}^n \leq \mathcal{A}$ for some $n \in \mathbb{N}$.*

Proof. The inclusion of supports says exactly that $\mathcal{B}$ is contained in the radical of $\mathcal{A}$. On an affine open, $\mathcal{B}$ is a finitely generated ideal inside the radical of $\mathcal{A}$, so some power of it lies in $\mathcal{A}$ there. Quasi-compactness bounds these exponents: the maximum over a finite affine cover works globally, containment of ideal sheaves being checkable on an affine cover. □

5 Blow-ups and admissible centres

Auxiliary mathematical input used here. For a quasi-coherent ideal sheaf $\mathcal{I}$ on a scheme X, the blow-up $\pi_{\mathcal{I}}\colon \mathrm{Bl}_{\mathcal{I}}(X) \to X$ exists; it is proper when $\mathcal{I}$ is of finite type; the inverse image ideal sheaf $\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_{\mathcal{I}}(X)}$ is invertible (*principalization*); and $\pi_{\mathcal{I}}$ restricts to an isomorphism over the open complement $V(\mathcal{I})^c$ of the centre.

Definition 5.1 (Admissible centre)**.** Let X be a scheme and $U \subseteq X$ an open. A *U-admissible centre* is a quasi-coherent ideal sheaf $\mathcal{I}$ on X which is of finite type and whose support is disjoint from U; equivalently, $\mathcal{I}$ restricts to the unit ideal on U. The associated *U-admissible blow-up* is $\pi_{\mathcal{I}} \colon \mathrm{Bl}_{\mathcal{I}}(X) \to X$. We write $\mathrm{pr}(\mathcal{I}) = V(\mathcal{I})^c$ for the *preserved open*; it contains U.

Proposition 5.2. *A U-admissible blow-up is proper, and it restricts to an isomorphism over the preserved open, in particular over U.*

Theorem 5.3 (Universal property of blowing up)**.** *Let $\mathcal{I}$ be a quasi-coherent ideal sheaf on X and let $t \colon T \to X$ be a morphism such that $\mathcal{I} \cdot \mathcal{O}_T$ is invertible. Then t factors through $\pi_{\mathcal{I}}$, and the factorization is unique: two morphisms $T \to \mathrm{Bl}_{\mathcal{I}}(X)$ over X agree as soon as $\mathcal{I} \cdot \mathcal{O}_T$ is invertible.*

Sketch. The statement is local on the charts of the blow-up, where it reduces to the universal property of the chart ring $A[\mathcal{I}/a]$: for a ring map $\varphi \colon A \to R$ with $\varphi(a)$ a nonzerodivisor dividing the extension of $\mathcal{I}$, the assignment $x/a^n \mapsto$ the unique solution of $y\varphi(a)^n = \varphi(x)$ is a well-defined ring map, and ring maps out of a chart ring which agree on the base and have regular value on the chart denominator coincide. Uniqueness globalizes through the localization describing the overlap of two charts. □

Theorem 5.4 (Blowing up a product)**.** *For quasi-coherent ideal sheaves $\mathcal{I}, \mathcal{L}$ on X there is an isomorphism over X*

$$\mathrm{Bl}_{\mathcal{I} \cdot \mathcal{L}}(X) \;\cong\; \mathrm{Bl}_{\mathcal{L} \cdot \mathcal{O}_{\mathrm{Bl}_{\mathcal{I}}(X)}}(\mathrm{Bl}_{\mathcal{I}}(X)).$$

Moreover multiplying a centre by an invertible ideal sheaf does not change the blow-up.

Sketch. Both statements follow from Theorem 5.3. On the iterated blow-up the inverse image of $\mathcal{I} \cdot \mathcal{L}$ is invertible, which produces a morphism in one direction; on $\mathrm{Bl}_{\mathcal{I} \cdot \mathcal{L}}(X)$ the inverse images of $\mathcal{I}$ and of $\mathcal{L}$ are invertible, because a factor of an invertible product ideal is invertible, which produces lifts in the other direction; the two composites are identities by uniqueness of lifts. □

Theorem 5.5 (Finite-degree approximation on a blow-up)**.** *Let X be quasi-compact quasi-separated, let $\mathcal{I}$ be a finite-type centre on X and let $\mathcal{J}$ be a finite-type ideal sheaf on $\mathrm{Bl}_{\mathcal{I}}(X)$. Then there are an integer $d \geq 1$ and a finite-type ideal sheaf $\mathcal{L} \leq \mathcal{I}^d$ on X with*

$$\mathcal{L} \cdot \mathcal{O}_{\mathrm{Bl}_{\mathcal{I}}(X)} \;=\; \left(\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_{\mathcal{I}}(X)}\right)^d \cdot \mathcal{J}.$$

Sketch. The argument is organised around the maximal degree-d candidate. Over an affine open of the base, the blow-up is covered by the charts at the degree-one elements of a finite generating set of the centre; each generator of the chart ideal of $\mathcal{J}$ is cleared into a numerator lying in the d-th power of the centre and having the prescribed fraction in *every* chart of the same algebra. A finite-type subsheaf through these finitely many numerators is then extracted by Theorem 4.3, and it generates the required product chart by chart. □

Theorem 5.6 (Composition of blow-ups)**.** *Let X be quasi-compact quasi-separated, let $\mathcal{I}$ be a finite-type centre on X and let $\mathcal{J}$ be a finite-type centre on $\mathrm{Bl}_{\mathcal{I}}(X)$. Then there is a finite-type centre $\mathcal{K}$ on X with*

$$V(\mathcal{K}) = V(\mathcal{I}) \cup \pi_{\mathcal{I}}(V(\mathcal{J}))$$

and an isomorphism $\mathrm{Bl}_{\mathcal{K}}(X) \cong \mathrm{Bl}_{\mathcal{J}}(\mathrm{Bl}_{\mathcal{I}}(X))$ compatible with the structure morphisms to X.

Sketch. Let $\mathcal{L}$ be the finite-degree descent of $\mathcal{J}$ of Theorem 5.5 and put $\mathcal{K} = \mathcal{I} \cdot \mathcal{L}$. Theorem 5.4 identifies $\mathrm{Bl}_{\mathcal{K}}(X)$ with the blow-up of $\mathcal{L} \cdot \mathcal{O}_{\mathrm{Bl}_{\mathcal{I}}(X)}$; removing from the latter its invertible exceptional factor, again by Theorem 5.4, identifies it with $\mathrm{Bl}_{\mathcal{J}}(\mathrm{Bl}_{\mathcal{I}}(X))$. The support formula is read off from the same descent. □

Corollary 5.7 (Composition of admissible blow-ups)**.** *Let X be quasi-compact quasi-separated and $U \subseteq X$ an open. The composite of a U-admissible blow-up $\pi_1 \colon X_1 \to X$ with a $\pi_1^{-1}U$-admissible blow-up $X_2 \to X_1$ is again a U-admissible blow-up of X: there are a U-admissible centre $\mathcal{K}$ on X and an isomorphism $X_2 \cong \mathrm{Bl}_{\mathcal{K}}(X)$ under which the composite becomes $\pi_{\mathcal{K}}$.*

Theorem 5.8 (Extension of admissible centres across a quasi-compact open)**.** *Let X be quasi-compact quasi-separated, let $W \subseteq X$ be a quasi-compact open, let $U \leq W$ be an open and let $\mathcal{J}$ be a finite-type centre on W whose support is disjoint from U. Then there is a U-admissible centre $\mathcal{I}$ on X together with a morphism $\mathrm{Bl}_{\mathcal{J}}(W) \to \mathrm{Bl}_{\mathcal{I}}(X)$ over $W \to X$ making the square cartesian; that is, the blow-up of $\mathcal{I}$ restricts over W to the blow-up of $\mathcal{J}$.*

Proof. Extend $\mathcal{J}$ to a finite-type ideal sheaf on X by Theorem 4.3 and multiply it by a finite-type ideal cutting out the boundary of W (Theorem 4.6); the product is U-admissible, and blow-up formation commutes with restriction to an open subscheme. □

Lemma 5.9 (Blow-ups restrict to open subschemes)**.** *For an open immersion $O \to X$ and a quasi-coherent ideal sheaf $\mathcal{I}$ on X, the blow-up of $\mathcal{I}|_O$ is the restriction of $\mathrm{Bl}_{\mathcal{I}}(X)$ over O, the comparison square being cartesian. Consequently the canonical lift $\mathrm{Bl}_{\mathcal{I}\cdot\mathcal{O}_Y}(Y) \to \mathrm{Bl}_{\mathcal{I}}(X)$ attached to a morphism $Y \to X$ localizes on the base, and two centres agreeing over an open have isomorphic lifts there.*

5.1 Density along admissible blow-ups

Theorem 5.10 (Density is preserved)**.** *Let $\mathcal{I}$ be a quasi-coherent ideal sheaf on X.*

(i) *If $\mathcal{I}$ is invertible, the open complement of its support is scheme-theoretically dense.*

(ii) *The inverse image of $V(\mathcal{I})^c$ in $\mathrm{Bl}_{\mathcal{I}}(X)$ is scheme-theoretically dense, the inverse image of the centre being invertible.*

(iii) *If U is a quasi-compact scheme-theoretically dense open of X, its inverse image under a U-admissible blow-up is scheme-theoretically dense.*

(iv) *The inverse image of a dense subset of a quasi-compact quasi-separated scheme under an admissible blow-up is dense.*

Theorem 5.11 (Making the preserved open schematically dense)**.** *Let X be quasi-compact quasi-separated and $U \subseteq X$ a quasi-compact open. Then there is a U-admissible blow-up of X in which the inverse image of U is scheme-theoretically dense; one blows up a finite-type centre supported exactly on $X \setminus U$ (Theorem 4.6). The same holds dominating a prescribed U-admissible blow-up B: apply the absolute form on the total space of B and compose the two blow-ups (Corollary 5.7).*

5.2 Separating boundary pieces

Definition 5.12 (Separation blow-up)**.** For quasi-coherent ideal sheaves $\mathcal{I}_1, \mathcal{I}_2$ on X, the *separation blow-up* is the blow-up of the sum $\mathcal{I}_1 \sqcup \mathcal{I}_2$.

Theorem 5.13 (Separation of two closed subschemes)**.** *Let X be quasi-compact quasi-separated and let $\mathcal{I}_1, \mathcal{I}_2$ be finite-type quasi-coherent ideal sheaves. Then the strict transforms of the closed subschemes $V(\mathcal{I}_1)$ and $V(\mathcal{I}_2)$ along the separation blow-up of $\mathcal{I}_1 \sqcup \mathcal{I}_2$ have disjoint ranges.*

Sketch. The charts of $\mathrm{Bl}_{\mathcal{I}_1 \sqcup \mathcal{I}_2}(X)$ at degree-one elements coming from local sections of $\mathcal{I}_1$ or of $\mathcal{I}_2$ cover the blow-up. On a chart attached to a section of $\mathcal{I}_i$ the extended centre is generated by that section, so the strict transform of $V(\mathcal{I}_i)$ misses that chart entirely; hence the two strict transforms cannot meet. □

Theorem 5.14 (Preparation of a Cartier boundary)**.** *Let X be quasi-compact quasi-separated and $U \subseteq X$ a quasi-compact open. Then there is a U-admissible blow-up $\pi\colon X' \to X$ and an invertible ideal sheaf $\mathcal{J}$ on X' whose support is exactly the complement of $\pi^{-1}U$. On a locally Noetherian scheme the same holds for an arbitrary open U.*

6 Strict transforms

Definition 6.1 (Strict transform)**.** Let $f\colon Y \to X$ and $\pi\colon X' \to X$ be morphisms and let $U \subseteq X$ be an open. Inside the fibre product $Y \times_X X'$ consider the open locus lying over U,

$$L \;=\; (\text{the preimage of } U \text{ under } Y \times_X X' \to X).$$

The *strict transform* $\mathrm{St}_U(f,\pi)$ of f along π preserving U is the scheme-theoretic image of L in $Y \times_X X'$. It is a closed subscheme of $Y \times_X X'$; we write $\mathrm{St}_U(f,\pi) \to X'$ and $\mathrm{St}_U(f,\pi) \to Y$ for the two projections. For $\pi = \pi_{\mathcal{I}}$ the blow-up of an ideal sheaf $\mathcal{I}$ and $U = V(\mathcal{I})^c$ this is the classical strict transform.

Proposition 6.2 (Inherited properties and ranges)**.** *Separatedness, quasi-compactness, being locally of finite type and being a closed immersion pass from f to the projection $\mathrm{St}_U(f,\pi) \to X'$. The strict transform contains the whole open locus over U. Its underlying set is squeezed between the inverse image of the U-part of the image of f and the full inverse image of the image of f; consequently, disjointness of two closed subschemes of X persists for their strict transforms.*

Lemma 6.3 (Behaviour over the preserved open)**.** *If π is an isomorphism over U, then the open locus L is canonically $Y \times_X U$, hence identified with $f^{-1}U$, and the part of the strict transform lying over U is the ordinary base change $Y \times_X U$.*

Lemma 6.4 (A kernel criterion)**.** *Assume the inclusion of the open locus L is quasi-compact. Then the strict transform and its open locus have the same scheme-theoretic image in X'. Moreover, a morphism to $Y \times_X X'$ whose first component is an isomorphism over $f^{-1}U$ on a schematically dense preimage of $f^{-1}U$ cuts out the same closed subscheme of $Y \times_X X'$ as the open locus over U.*

Theorem 6.5 (The strict transform is a blow-up)**.** *Let $f\colon Y \to X$ be a morphism and $\mathcal{I}$ a quasi-coherent ideal sheaf on X. Then there is an isomorphism*

$$\mathrm{Bl}_{\mathcal{I}\cdot\mathcal{O}_Y}(Y) \;\xrightarrow{\sim}\; \mathrm{St}_{V(\mathcal{I})^c}(f,\pi_{\mathcal{I}})$$

compatible with the projections to Y and with the structure morphisms to X. No finiteness or quasi-compactness hypotheses are required.

Sketch. Consider the comparison morphism $\psi\colon \mathrm{Bl}_{\mathcal{I}\cdot\mathcal{O}_Y}(Y) \to Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$ whose components are the structure morphism of $\mathrm{Bl}_{\mathcal{I}\cdot\mathcal{O}_Y}(Y)$ and the lift supplied by Theorem 5.3. It is a closed immersion: the assertion is local, and over a chart of $\mathrm{Bl}_{\mathcal{I}}(X)$ at a degree-one element x it is the spectrum of the canonical ring map $B \otimes_A A[\mathcal{I}/x] \to B[\mathcal{I}B/\varphi(x)]$, which is surjective because a fraction $y/\varphi(a)^n$ with $y \in \mathcal{I}^n B$ decomposes as a B-linear combination of fractions coming from the chart ring of $\mathrm{Bl}_{\mathcal{I}}(X)$. Furthermore ψ has the same kernel as the inclusion of the open locus: the open locus factors through ψ as the preimage of $V(\mathcal{I}\cdot\mathcal{O}_Y)^c$ in $\mathrm{Bl}_{\mathcal{I}\cdot\mathcal{O}_Y}(Y)$, which is schematically dense there because the exceptional ideal is invertible (Theorem 5.10(ii)); so precomposing with it does not change the kernel, and Lemma 6.4 applies. Since the strict transform is by definition the closed subscheme cut out by the kernel of the open locus, the two closed subschemes of $Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$ coincide. □

Corollary 6.6 (Blowing up a closed subscheme)**.** *Let $j\colon Z \to X$ be a closed immersion and $\mathcal{I}$ a quasi-coherent ideal sheaf on X. Then $\mathrm{Bl}_{\mathcal{I}\cdot\mathcal{O}_Z}(Z)$ is a closed subscheme of $\mathrm{Bl}_{\mathcal{I}}(X)$, compatibly with the two blow-up morphisms.*

7 Elimination of indeterminacy after Deligne and Conrad

Throughout this section, X is a quasi-compact quasi-separated scheme and $U \leq V$ are quasi-compact opens of X. We study the problem of extending a *partial section*: a morphism $f\colon U \to Y$ with $p \circ f = \iota_U$ for a morphism $p\colon Y \to X$. The aim is to extend f, after an admissible blow-up of the base, to a section over an open neighbourhood of the preserved open, with *closed graph* in the fibre product.

7.1 Deligne's disjointness blow-up

Theorem 7.1 (Disjointness blow-up)**.** *Let $p\colon Z \to X$ be a morphism, let $f\colon Y \to Z$ be a morphism over X whose source lies over an open $U \leq V$ of X, and let $\mathcal{I}_1, \mathcal{I}_2$ be quasi-coherent ideal sheaves on X such that*

(a) *$V \setminus U \subseteq V(\mathcal{I}_1)$ as sets, so that $T_1 = V(\mathcal{I}_1^2)$ is a square-zero thickening of a closed subscheme containing the boundary of U inside V;*

(b) *$V(\mathcal{I}_2) \subseteq X \setminus V$ as sets;*

(c) *the closed subscheme $p^{-1}T_1 \cap \overline{Y}$ of Z factors through $p^{-1}T_2$, where $\overline{Y}$ is the scheme-theoretic image of f; in ideal-sheaf language, $\mathcal{I}_2 \cdot \mathcal{O}_Z \leq (\mathcal{I}_1^2) \cdot \mathcal{O}_Z \sqcup \ker f$.*

Blow up $\mathcal{I}_1 \sqcup \mathcal{I}_2$ and base change everything along the blow-up morphism q. Then the scheme-theoretic image of the base change of f no longer meets the part of the base change of Z lying over $\overline{q^{-1}(V \setminus U)}$.

Sketch. The charts of $\mathrm{Bl}_{\mathcal{I}_1 \sqcup \mathcal{I}_2}(X)$ at sections of $\mathcal{I}_1$ are harmless: there the inverse images of $\mathcal{I}_1$ and of $\mathcal{I}_1 \sqcup \mathcal{I}_2$ have the same support, so a point of such a chart lying over $V(\mathcal{I}_1)$ lies over $V(\mathcal{I}_2)$ as well, which is disjoint from V; hence such a chart misses $q^{-1}(V \setminus U)$ altogether, and therefore also its closure. Only the charts at sections s_2 of $\mathcal{I}_2$ remain. On an affine probe of the base change lying over such a chart, the extended centre is generated by the image σ of s_2, so the factorization hypothesis reads $\sigma \in \sigma a_1 + \mathfrak{n}$, with a_1 the extension of $\mathcal{I}_1$ and $\mathfrak{n}$ the kernel of the base change of f; that is, $\sigma(1-h) \in \mathfrak{n}$ with $h \in a_1$. Since the source of f lies over U, which misses $V(\mathcal{I}_1 \sqcup \mathcal{I}_2)$, the extension of the centre there is the unit ideal, so multiplying by σ does not change the kernel and $1 - h \in \mathfrak{n}$. Then $1 = h + (1-h)$ shows that the inverse images of $\mathcal{I}_1$ and the kernel generate the unit ideal on the probe, so their supports do not meet. □

7.2 The case of a target that is quasi-affine over the base

Theorem 7.2 (Extension into a relative affine space)**.** *Let X be affine, let $U \subseteq X$ be a quasi-compact open, let σ be finite and let $f\colon U \to \mathbb{A}(\sigma; X)$ be a morphism over X. Then there are a U-admissible centre $\mathcal{I}$ on X, an open $W \subseteq \mathrm{Bl}_{\mathcal{I}}(X)$ containing $\pi_{\mathcal{I}}^{-1}U$ and a morphism $g\colon W \to \mathbb{A}(\sigma; X)$ over X which agrees with f over $\pi_{\mathcal{I}}^{-1}U$, such that the graph $W \to \mathrm{Bl}_{\mathcal{I}}(X) \times_X \mathbb{A}(\sigma; X)$ has closed range.*

Sketch. Cover U by finitely many basic opens $D(u)$. On $D(u)$ the coordinates of f are fractions $g_{u,j}/u^N$ for a single large N, and after one further enlargement of N the compatibility identities $g_{u,j}\, u'^N = g_{u',j}\, u^N$ hold exactly in $\Gamma(X, \top)$. Let $\mathcal{I}$ be generated by all u^N and all $g_{u,j}$; it restricts to the unit ideal on U, hence is U-admissible. On the blow-up chart at u^N the j-th coordinate extends as the degree-one fraction $(g_{u,j} \cdot x)/(u^N \cdot x)$, and the exact identities make the chart transitions hold on the nose, so the local extensions glue over the union W of the charts at the u^N, which contains $\pi_{\mathcal{I}}^{-1}U$. Closedness of the graph is checked on the chart at each generator a of $\mathcal{I}$: there the graph is the vanishing locus of the sections $\hat{v}_i t_j - \hat{w}_{ij}$ built from the fractions $v_i = (u_i^N x)/(ax)$ and $w_{ij} = (g_{u_i,j} x)/(ax)$, because these fractions generate the unit ideal on the chart. □

Theorem 7.3 (Case 1: quasi-affine target over an affine base)**.** *Let X be affine, let $U \subseteq X$ be a quasi-compact open, let Y be a quasi-affine scheme and let $p\colon Y \to X$ be locally of finite type. Let $f\colon U \to Y$ satisfy $p \circ f = \iota_U$. Then there are a U-admissible centre $\mathcal{I}$ on X, an open $W \subseteq \mathrm{Bl}_{\mathcal{I}}(X)$ containing $\pi_{\mathcal{I}}^{-1}U$ and a morphism $g\colon W \to Y$ with $p \circ g = \pi_{\mathcal{I}}|_W$, agreeing with f over $\pi_{\mathcal{I}}^{-1}U$, such that the graph $W \to Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$ has closed range and $\pi_{\mathcal{I}}^{-1}U$ is scheme-theoretically dense in $\mathrm{Bl}_{\mathcal{I}}(X)$.*

Proof. Embed Y into a relative affine space over X by Theorem 3.5, extend f composed with that immersion by Theorem 7.2, and then compose with one further admissible blow-up making the preserved open schematically dense (Theorem 5.11), transporting the extension along the comparison of the two blow-ups. □

Theorem 7.4 (Case 2: quasi-affine target over a quasi-compact quasi-separated base, $U = V$)**.** *Let X be quasi-compact quasi-separated, let $U \subseteq X$ be a quasi-compact open, and let $p\colon Y \to X$ be separated, locally of finite type and quasi-affine as a morphism. Let $f\colon U \to Y$ satisfy $p \circ f = \iota_U$. Then the conclusion of Theorem 7.3 holds verbatim.*

Sketch. Apply Case 1 over each member of a finite affine open cover of X, extend the local centres found there to the whole base (Theorem 4.3), and blow up the product of the extensions times a finite-type ideal cutting out $X \setminus U$ (Theorem 4.6). The support of the resulting centre is exactly $X \setminus U$, so the inverse image of U is scheme-theoretically dense and in particular topologically dense. That density makes the local extension loci pairwise compatible, which in turn makes the graph of the glued extension meet the part of $Y \times_X X'$ over each member of the cover in the closed local graph there. □

Theorem 7.5 (Case 3: two nested opens)**.** *Let X be quasi-compact quasi-separated, let $U \leq V$ be quasi-compact opens of X, let Y be quasi-compact quasi-separated and let $p\colon Y \to X$ be separated, locally of finite type and quasi-affine as a morphism. Let $f\colon U \to Y$ satisfy $p \circ f = \iota_U$ and assume the range of f is closed in $p^{-1}V$. Then there are a V-admissible centre $\mathcal{I}$ on X, an open $W \subseteq \mathrm{Bl}_{\mathcal{I}}(X)$ containing $\pi_{\mathcal{I}}^{-1}U$, and a morphism $g\colon W \to Y$ with $p \circ g = \pi_{\mathcal{I}}|_W$ agreeing with f over $\pi_{\mathcal{I}}^{-1}U$, whose graph $W \to Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$ has closed range. Density of U in X is not assumed.*

Sketch. Two opening reductions bring the situation to Case 2. First, one V-admissible blow-up $\pi_1\colon X_1 \to X$ produced by Theorem 7.1 makes the closure of the image of the base change of f disjoint from the part lying over $\overline{V \setminus U}$; here the two centres are a finite-type subideal $\mathcal{I}_1$ of the vanishing ideal of the reduced $\overline{V \setminus U}$, supplied by Theorem 4.5(i), and $\mathcal{I}_2 = \mathcal{I}_1^2 \sqcup \mathcal{K}^n$, where $\mathcal{K}$ cuts out $X \setminus V$ and n is large enough by Lemma 4.7. Second, since the carrier of a scheme is prespectral, a quasi-compact subset of an open is contained in a quasi-compact open of it; this yields a quasi-compact open $X_0 \subseteq X_1$ containing that closure's image and meeting $\pi_1^{-1}V$ exactly in $\pi_1^{-1}U$. Over X_0 the two opens coincide, so Case 2 applies to the base change of p; the centre found over X_0 is extended to X_1 by Theorem 5.8, whose hypothesis is exactly $X_0 \cap \pi_1^{-1}V = \pi_1^{-1}U$, and the two admissible blow-ups are composed by Corollary 5.7. □

7.3 The general case

Theorem 7.6 (The closed graph is quasi-finite over the blow-up)**.** *Let X be quasi-compact quasi-separated, let $U \leq V$ be quasi-compact opens, let Y be quasi-compact quasi-separated and let $p\colon Y \to X$ be separated and locally of finite type. Let $f\colon U \to Y$ satisfy $p \circ f = \iota_U$ and have range closed in $p^{-1}V$. Then there are a V-admissible centre $\mathcal{I}$ on X and an open immersion $\sigma\colon U \to \mathrm{Bl}_{\mathcal{I}}(X)$ lifting ι_U, such that for the graph $F = (f, \sigma)\colon U \to Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$ the scheme-theoretic image $\operatorname{im} F$ is quasi-finite, of finite type, separated and quasi-compact over $\mathrm{Bl}_{\mathcal{I}}(X)$.*

Sketch. Cover Y by finitely many affine opens Z. For each of them the composite $Z \to Y \to X$ is quasi-affine as a morphism, so Theorem 7.5 applies to the restriction of f to $f^{-1}Z$ and produces

a V-admissible centre $\mathcal{I}_Z$ on X together with an open W_Z of $\mathrm{Bl}_{\mathcal{I}_Z}(X)$ carrying an extension of $f|_{f^{-1}Z}$ with closed graph in $Z \times_X \mathrm{Bl}_{\mathcal{I}_Z}(X)$. Let $\mathcal{I} = \prod_Z \mathcal{I}_Z$. Each $\mathcal{I}_Z$ pulls back to an invertible ideal on $\mathrm{Bl}_{\mathcal{I}}(X)$, so Theorem 5.3 provides comparison morphisms $\mathrm{Bl}_{\mathcal{I}}(X) \to \mathrm{Bl}_{\mathcal{I}_Z}(X)$ over X, and the closed graph of each local extension pulls back to a closed subset of the open $\mathrm{pr}_Y^{-1}Z$ of $Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$ which is a section over an open of $\mathrm{Bl}_{\mathcal{I}}(X)$. These opens cover $Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$, and the closure of the range of F meets each of them inside the corresponding section; hence the fibres of $\operatorname{im} F \to \mathrm{Bl}_{\mathcal{I}}(X)$ are finite. □

Auxiliary mathematical input used here. Zariski's Main Theorem. A separated, quasi-compact, locally quasi-finite morphism of finite type factors as an open immersion into the relative normalization of the target in the source followed by an integral, in particular affine, morphism. Consequently such a morphism is quasi-affine as a morphism, with no hypothesis on the target, and its source is a quasi-affine scheme whenever the target is.

Theorem 7.7 (Extension of a partial section)**.** *Let X be quasi-compact quasi-separated, let $U \leq V$ be quasi-compact opens of X with U dense in X, let Y be quasi-compact quasi-separated and let $p\colon Y \to X$ be separated, quasi-compact and locally of finite type. Let $f\colon U \to Y$ satisfy $p \circ f = \iota_U$ and have range closed in $p^{-1}V$. Then there are a V-admissible centre $\mathcal{I}$ on X, an open $W \subseteq \mathrm{Bl}_{\mathcal{I}}(X)$ containing $\pi_{\mathcal{I}}^{-1}U$ and a morphism $g\colon W \to Y$ with $p \circ g = \pi_{\mathcal{I}}|_W$ which agrees with f over $\pi_{\mathcal{I}}^{-1}U$, such that the graph $W \to Y \times_X \mathrm{Bl}_{\mathcal{I}}(X)$ has closed range.*

Proof. If p is quasi-affine as a morphism this is Theorem 7.5. In general, Theorem 7.6 produces a V-admissible blow-up $q\colon \widetilde{X} \to X$ over which the scheme-theoretic image of the graph is quasi-finite, of finite type, separated and quasi-compact over $\widetilde{X}$; Zariski's Main Theorem turns these four properties into quasi-affineness of $\operatorname{im} F \to \widetilde{X}$ *as a morphism* (neither $\widetilde{X}$ nor $\operatorname{im} F$ need be quasi-affine as a scheme). Two purely formal reductions finish the proof: one may base change the whole situation along a V-admissible blow-up without leaving the statement, the blow-up produced downstairs being the composite of Corollary 5.7; and one may replace the target by a closed subscheme through which the section factors, here by the scheme-theoretic image of U. After these two reductions the target is quasi-affine over the base and Theorem 7.5 applies. □

Corollary 7.8 (The case $V = U$)**.** *Let X be quasi-compact quasi-separated, let $U \subseteq X$ be a dense quasi-compact open, let Y be quasi-compact quasi-separated and let $p\colon Y \to X$ be separated, quasi-compact and locally of finite type. Then any $f\colon U \to Y$ with $p \circ f = \iota_U$ extends, after a U-admissible blow-up, to a section over an open of the blow-up containing the preimage of U, with closed graph. No closedness hypothesis is needed: over U the morphism f is a section of the separated morphism $p|_U$, hence a closed immersion, so its range is closed in $p^{-1}U$.*

8 Modifications

Theorem 8.1 (Elimination of indeterminacy)**.** *Let X be quasi-compact quasi-separated, let $U \subseteq X$ be a quasi-compact open and let $q\colon Z \to X$ be separated, locally of finite type and quasi-compact. Assume q restricts to an isomorphism over U. Then there is a U-admissible centre $\mathcal{I}$ on X such that the strict transform of q along $\pi_{\mathcal{I}}$, preserving $\mathrm{pr}(\mathcal{I})$, is an open immersion into $\mathrm{Bl}_{\mathcal{I}}(X)$.*

Sketch. First prepare the boundary: blow up a finite-type ideal whose support is exactly $X \setminus U$ (Theorem 4.6), so that U becomes scheme-theoretically dense (Theorem 5.10). Next extend the inverse of q over U to a section over an open W of a second admissible blow-up, by Corollary 7.8. Compose the two blow-ups (Corollary 5.7). Finally, the graph of the extension is closed and

contains the schematically dense U-graph, so it cuts out the strict transform, which is therefore isomorphic to the open W. □

Remark 8.2. The argument just sketched replaces the classical derivation of Theorem 8.1 from module flattening. The surveyed material contains no flattening theorem; see §13.

Theorem 8.3 (Transport of scheme-theoretic images)**.** *Let $\mathcal{I}$ be a U-admissible centre on X. Let h be a quasi-compact morphism into a fibre product over X whose first component lands in U, and let h' be a lift of h along the base change of $\pi_{\mathcal{I}}$. Then the scheme-theoretic image of h' is the strict transform of the scheme-theoretic image of h: the kernel of h' equals the kernel of the projection to the blown-up fibre product of the strict transform of $\iota_{\operatorname{im} h}$ along the base change of $\pi_{\mathcal{I}}$, preserving the preimage of the preserved open.*

Definition 8.4 (Common modification)**.** Let $s_1\colon X_1 \to S$, $s_2\colon X_2 \to S$ be morphisms and let $j_1\colon U \to X_1$, $j_2\colon U \to X_2$ be open immersions with $s_1 \circ j_1 = s_2 \circ j_2$. A *common modification* consists of $j_i(U)$-admissible centres $\mathcal{I}_i$ on X_i, a scheme W with a morphism $w\colon W \to S$, open immersions $k_i\colon \operatorname{Bl}_{\mathcal{I}_i}(X_i) \to W$ over S whose ranges cover W, and lifts $u_i\colon U \to \operatorname{Bl}_{\mathcal{I}_i}(X_i)$ of j_i with $k_1 \circ u_1 = k_2 \circ u_2$, such that w is separated, locally of finite type and quasi-compact.

Theorem 8.5 (Existence of common modifications)**.** *Let S be quasi-compact quasi-separated, let $s_1\colon X_1 \to S$ and $s_2\colon X_2 \to S$ be separated, locally of finite type and quasi-compact, and let $j_1\colon U \to X_1$, $j_2\colon U \to X_2$ be quasi-compact open immersions with $s_1 \circ j_1 = s_2 \circ j_2$. Then a common modification of (s_1, s_2, j_1, j_2) exists.*

Sketch. Form the graph $\gamma = (j_1, j_2)\colon U \to X_1 \times_S X_2$, let X_{12} be its scheme-theoretic image and let p_1, p_2 be the two projections of X_{12}. Both are isomorphisms over U, so Theorem 8.1 applies to each: after admissible blow-ups of X_1 and X_2, the strict transforms of p_1 and p_2 become open immersions. By Theorem 6.5 these strict transforms are blow-ups of X_{12} at the pulled-back centres; the product of the two pulled-back centres dominates both, and by Theorem 5.8 and Corollary 5.7 the resulting blow-up of X_{12} sits, as an open subscheme, inside a single admissible blow-up of each X_i. Gluing the two blow-ups along that common open produces W, which is separated, locally of finite type and quasi-compact over S. □

Definition 8.6 (Graph extension)**.** Let $u\colon U \to S$ be a morphism, $V \subseteq U$ a quasi-compact open and $c = (Z, g, i)$ a compactification of $u \circ \iota_V$. A *graph extension* of c consists of an $i(V)$-admissible centre on Z, an open V^{gr} of the blow-up and a proper morphism $\psi\colon V^{\mathrm{gr}} \to U$ over S extending the inclusion of V, such that V embeds in V^{gr} as exactly the preimage $\psi^{-1}V$ and ψ is an isomorphism over V.

Theorem 8.7 (Existence of graph extensions)**.** *Let S be quasi-compact quasi-separated, let $u\colon U \to S$ be separated, locally of finite type and quasi-compact, let $V \subseteq U$ be a quasi-compact open and let c be a compactification of $u \circ \iota_V$. Then a graph extension of c exists.*

Sketch. Take the graph $\gamma = (i, \iota_V)\colon V \to Z \times_S U$, a partial section of the first projection over the open immersion i. Its scheme-theoretic image is transported through an admissible blow-up by Theorem 8.3, and Theorem 8.1 makes the corresponding strict transform an open immersion; V^{gr} is that open, and ψ is the second projection, proper because $Z \times_S U \to U$ is proper. □

Theorem 8.8 (Isomorphism near a boundary closure)**.** *Let X be quasi-compact quasi-separated, let $U \subseteq X$ be a quasi-compact open and let $q\colon Z \to X$ be proper. Let $T \subseteq U$ be a subset which is closed in U and let $N \subseteq X$ be an open containing T over which q is an isomorphism. Then there are a U-admissible centre $\mathcal{I}$ on X and an open $N' \subseteq \operatorname{Bl}_{\mathcal{I}}(X)$ with $\overline{\pi_{\mathcal{I}}^{-1}T} \subseteq N'$ such that the strict transform of q along $\pi_{\mathcal{I}}$ is an isomorphism over N'.*

9 The two-open compactification theorem

Throughout this section S is Noetherian and $u\colon U \to S$ is separated, locally of finite type and quasi-compact. Let O, O' be opens of U.

9.1 Prepared sides

Definition 9.1 (A side)**.** A *side* for (u, O, O') consists of a proper morphism $s\colon X \to S$, an open $V \subseteq X$, a proper morphism $\psi\colon V \to U$ over S, and an open immersion $\upsilon\colon O \to V$ with $\psi \circ \upsilon = \iota_O$, such that the range of υ is exactly $\psi^{-1}O$, ψ is an isomorphism over O, and the composite $O \to V \to X$ has dense range.

Definition 9.2 (Boundary loci; preparedness)**.** For a side σ put

$$\overline{Z}_{\mathrm{self}} = \overline{\iota_V\big(\psi^{-1}(U \setminus O')\big)}, \qquad \overline{Z}_{\mathrm{forb}} = \overline{\iota_V\big(\psi^{-1}(U \setminus O)\big)},$$

the *self boundary* and the *forbidden boundary*, both closed in X. The side is *prepared* if $\overline{Z}_{\mathrm{self}} \cap \overline{Z}_{\mathrm{forb}} = \emptyset$. The open complement $X^\circ = X \setminus \overline{Z}_{\mathrm{forb}}$ is the *interior* of the side; it contains the embedded copy of O.

Proposition 9.3 (Transport along admissible blow-ups)**.** *A side may be transported along a V-admissible blow-up of its ambient scheme; boundary loci transport to their preimages, so preparedness persists.*

Auxiliary mathematical input used here. *(i)* Every compactification of a quasi-compact morphism may be replaced by one whose open immersion is scheme-theoretically dominant, namely by passing to the scheme-theoretic image of the open immersion inside the ambient scheme. *(ii)* **Separation of constructible boundaries.** If T_1, T_2 are disjoint closed subsets of a quasi-compact open U of a quasi-compact quasi-separated scheme X with quasi-compact complements in U, then after a U-admissible blow-up of X the closures of the inverse images of T_1 and T_2 are disjoint.

Theorem 9.4 (Existence of a prepared side)**.** *Let $O \sqcup O' = U$ and assume $u \circ \iota_O$ is compactifiable. Then a prepared side for (u, O, O') exists.*

Sketch. Choose a scheme-theoretically dense compactification of $u \circ \iota_O$, extend it by the graph construction of Theorem 8.7, and separate the two boundary closures by the constructible-boundary separation blow-up, which rests on Theorem 5.13. □

9.2 The comparison space

Let σ_1 be a side for (u, O_1, O_2) and σ_2 a side for (u, O_2, O_1), with ambient schemes X_1, X_2 and proper extensions ψ_1, ψ_2.

Definition 9.5 (Comparison space)**.** The *comparison morphism* is the canonical morphism $V_1 \times_U V_2 \to X_1 \times_S X_2$. The *comparison space* X_{12} is its scheme-theoretic image, with the two projections $\mathrm{pr}_1\colon X_{12} \to X_1$ and $\mathrm{pr}_2\colon X_{12} \to X_2$, both proper.

Definition 9.6 (Key boundary disjointness)**.** The pair (σ_1, σ_2) satisfies *key boundary disjointness* if no point of X_{12} projects into the forbidden boundary on both sides:

$$X_{12} \cap \big(\overline{Z}_{1,\mathrm{forb}} \times_S \overline{Z}_{2,\mathrm{forb}}\big) = \emptyset.$$

The *comparison boundary* is $\mathrm{pr}_1^{-1}\overline{Z}_{1,\mathrm{forb}}$; key disjointness says that its image under pr_2 avoids $\overline{Z}_{2,\mathrm{forb}}$.

Lemma 9.7 (Consequence for valuations)**.** *Key boundary disjointness implies that, for any pair of valuation-ring extensions of a common generic point of $O_1 \cap O_2$ to the two ambient models, at least one of the two avoids its forbidden boundary.*

Theorem 9.8 (The improved comparison)**.** *Given prepared sides σ_1, σ_2 with $O_1 \sqcup O_2 = U$, there is a V_1-admissible blow-up of X_1 after which the comparison boundary of the improved pair is contained in the preimage of the self boundary of σ_1.*

Sketch. Four ingredients enter. First, over the proper-extension open V_1 the comparison space is the fibre product $V_1 \times_U V_2$ itself: the comparison morphism is an immersion, being the base change of the diagonal of u followed by an open immersion, and it is proper over V_1, so its range is closed there. Second, the first comparison projection is an isomorphism over the image in X_1 of the locus of V_1 lying above O_2; indeed ψ_2 is an isomorphism over O_2 and, over V_1, the first projection is its base change. That open contains the forbidden boundary of σ_1, because $U \setminus O_1 \subseteq O_2$. Third, after a V_1-admissible blow-up the comparison space of the improved pair is a closed subscheme of the strict transform of the old comparison space, so the improved first projection is injective wherever the strict-transform projection is an isomorphism; this is where Theorem 8.8 enters. Fourth, a density criterion: controlling the second projection on a subset whose closure swallows the comparison boundary suffices, because the self boundary is itself a closure. □

Theorem 9.9 (Existence of a key-disjoint prepared pair)**.** *Let $O_1 \sqcup O_2 = U$ and assume both $u \circ \iota_{O_1}$ and $u \circ \iota_{O_2}$ are compactifiable. Then there exist prepared sides σ_1 for (u, O_1, O_2) and σ_2 for (u, O_2, O_1) satisfying key boundary disjointness.*

Proof. Take prepared sides on both sides by Theorem 9.4, improve the first by Theorem 9.8, and observe that a comparison boundary landing in the self boundary of a *prepared* second side cannot meet its forbidden boundary. □

9.3 The intermediate schemes and the theorem

Definition 9.10 (Intermediate scheme)**.** For a side σ with interior X°, the *intermediate scheme* is the pushout

$$W \;=\; U \;\sqcup_O\; X^\circ,$$

the gluing of U and X° along the embedded copy of O, with its two chart open immersions and its structure morphism $W \to S$.

Proposition 9.11 (Geometry of the intermediate scheme)**.** (i) *The overlap morphism $O \to U \times_S X^\circ$ is a closed immersion: its range is the preimage of the closed range of the proper graph $V \to U \times_S X$, and the range of the embedding of O in V is exactly $\psi^{-1}O$.*

(ii) *Consequently $W \to S$ is separated: a morphism out of a scheme covered by two opens is separated as soon as the overlap morphism into the product of the two charts over the base is a closed immersion.*

(iii) *The copy of U is dense in W.*

Auxiliary mathematical input used here. The dense-source valuative criterion. Let $f\colon X \to S$ be a morphism and $h\colon U \to X$ quasi-compact and dominant. If every valuative square for f whose generic point factors through h admits a unique lift, then f is universally closed; the corresponding statement for uniqueness gives separatedness. Moreover, for a scheme-theoretically dominant quasi-compact morphism and a point of its scheme-theoretic image, a valuation probe centred at that point exists.

Theorem 9.12 (The two-open compactification theorem)**.** *Let S be Noetherian and let $u\colon U \to S$ be separated, locally of finite type and quasi-compact. Suppose U is covered by two opens O_1, O_2 whose intersection $O_1 \cap O_2$ is dense in U, and suppose that both $u \circ \iota_{O_1}$ and $u \circ \iota_{O_2}$ admit compactifications. Then u admits a compactification.*

Sketch. Take a key-disjoint prepared pair by Theorem 9.9 and form the two intermediate schemes W_1, W_2 of Definition 9.10; by Proposition 9.11 they are separated, of finite type and Noetherian over S, and each contains a dense copy of U. Apply Theorem 8.5 to W_1, W_2 and their common copy of U; let W be the resulting common modification. Density of $O_1 \cap O_2$ transports through the finite modification diagram, by Theorem 5.10(iv), so $O_1 \cap O_2$ is dense in W. Every valuative square through this dense source lifts uniquely: extend the generic point to the two proper sides, pass to the side which avoids its forbidden boundary by Lemma 9.7, and lift along the proper admissible blow-up. The dense-source valuative criterion then shows that $W \to S$ is universally closed, hence proper, and $U \to W$ is an open immersion; this is a compactification of u. □

10 The Noetherian case

Theorem 10.1 (A dense affine cover)**.** *Let X be a quasi-compact scheme with finitely many irreducible components. Then X admits a finite affine open cover $U_0, \dots, U_{n-1}$ whose total intersection $\bigcap_i U_i$ contains every generic point of X, and is therefore dense in X.*

Sketch. Every point has an affine open neighbourhood containing all generic points: take the disjoint union of an affine neighbourhood of the point avoiding the components the point does not lie on, together with one small affine neighbourhood of each remaining generic point. Quasi-compactness extracts a finite cover from this family, and the total intersection of such a cover contains all generic points, hence is dense. □

Definition 10.2 (Partial unions)**.** For a finite family $U_0, \dots, U_{n-1}$ of opens of X put $X_r = \bigcup_{i<r} U_i$, so $X_0 = \emptyset$ and $X_n = \bigcup_i U_i$.

Lemma 10.3 (The gluing step)**.** *Let S be Noetherian and $f\colon X \to S$ separated, locally of finite type and quasi-compact. Let $D \subseteq X$ be dense and contained in two opens $V, W \subseteq X$, and assume both $f \circ \iota_V$ and $f \circ \iota_W$ are compactifiable. Then $f \circ \iota_{V \cup W}$ is compactifiable.*

Proof. Apply Theorem 9.12 to the two opens V, W of the scheme $V \cup W$ and to the restriction of f. Their intersection contains D, which is dense in $V \cup W$ because $V \cup W$ is open in X; the two restrictions inherit compactifiability from V and W by Proposition 2.3(iii); and $V \cup W$ is Noetherian, X being locally Noetherian and quasi-compact. □

Theorem 10.4 (Nagata compactification over a Noetherian base)**.** *Let S be a Noetherian scheme and let $f\colon X \to S$ be separated, locally of finite type and quasi-compact. Then f admits a compactification.*

Proof. Since f is locally of finite type over a Noetherian base and quasi-compact, X is a Noetherian scheme; in particular it has finitely many irreducible components, so Theorem 10.1 provides a finite affine open cover $U_0, \dots, U_{n-1}$ with dense total intersection D. We prove by induction on r that $f \circ \iota_{X_r}$ is compactifiable, for the partial unions of Definition 10.2.

For $r = 0$ the scheme X_0 is empty, so $X_0 \to S$ is proper and hence compactifiable by Proposition 2.3(i). For $r = 1$ the scheme $X_1 = U_0$ is affine, and Theorem 3.7 applies. For the step from $r \geq 1$ to $r+1$, write $X_{r+1} = X_r \cup U_r$. The dense set D is contained in $U_0 \subseteq X_r$ and in U_r, and both $X_r \to S$ and $U_r \to S$ are compactifiable, the first by induction and the second again by Theorem 3.7. Lemma 10.3 gives the claim for X_{r+1}. The hypotheses of the gluing step hold at every stage because separatedness, local finite type and quasi-compactness of $X_r \to S$ are inherited from f by the open subscheme X_r. Since $X_n = X$, the last stage compactifies f. □

11 Removing the Noetherian hypothesis

11.1 Absolute Noetherian approximation

Definition 11.1 (Noetherian approximation)**.** A *Noetherian approximation* of a scheme S consists of a small cofiltered index category I, a diagram $D\colon I \to \mathrm{Sch}$ whose values are of finite type over $\operatorname{Spec}\mathbb{Z}$ (that is, quasi-compact and locally of finite type over $\operatorname{Spec}\mathbb{Z}$) and whose transition morphisms are affine, together with a cone from S exhibiting S as the limit $\varprojlim_i D_i$. Every stage of such a system is a Noetherian scheme.

Theorem 11.2 (Absolute Noetherian approximation)**.** *Every scheme which is quasi-compact and quasi-separated admits a Noetherian approximation.*

Sketch. For an affine scheme S the spectra of the finitely generated $\mathbb{Z}$-subalgebras of $\Gamma(S, \top)$ form a Noetherian approximation: $\Gamma(S, \top)$ is the filtered colimit of those subalgebras, and Spec, being a right adjoint, turns that colimit into a limit. The general case is an induction over a finite affine cover, each step extending an approximation of a quasi-compact open U across one more affine open V. The stages of a step are indexed by triples consisting of a stage of the approximation of U, lifts of finitely many sections of V whose basic opens cover $U \cap V$, and a finitely generated $\mathbb{Z}$-subalgebra of a suitable fibre product of section rings large enough that the relevant overlap becomes a quasi-affine open of its spectrum. These stages form a cofiltered category and the functor remembering the underlying index is initial, so limits over the system of overlaps may be computed over the category of stages. The limit property is verified chart by chart, using that a cone over a cofiltered diagram of schemes is a limit cone as soon as its restrictions to the members of an open cover of one stage, and to their pairwise intersections, are limit cones. □

11.2 Descending finite presentation and separatedness

Theorem 11.3 (Descent of morphisms of finite presentation)**.** *Let (D_i) be a cofiltered inverse system of quasi-compact quasi-separated schemes with affine transition morphisms, and let X be of finite presentation over $\varprojlim_i D_i$. Then X is the base change of a scheme of finite presentation over one of the D_i.*

Sketch. The affine case is treated first: an affine X lying over an affine open of a stage descends after base changing the whole system to that open. The general case is an induction over a finite cover of X by such affine opens; at each step two finite-stage models are glued along a descent of their intersection. The gluing data comes from the facts that a quasi-compact open descends to a stage and that two finite-stage models of the same scheme become isomorphic at a later stage; the gluing itself is a pushout of two open immersions, and the glued morphism is again cartesian. □

Theorem 11.4 (Descent of separatedness)**.** *Let (D_i) be a cofiltered inverse system of quasi-compact quasi-separated schemes with affine transition morphisms. If $\varprojlim_i D_i$ is separated, then D_i is separated for every i past some stage. If all D_i lie over a fixed base S and $\varprojlim_i D_i \to S$ is separated, then $D_i \to S$ is separated for every i past some stage.*

Sketch. Both go through an affine criterion: a scheme is separated as soon as the affine opens of a finite affine cover pairwise have affine intersections and the sections over such an intersection are generated as a ring by the sections of the two members. Affineness of the intersections descends to a stage. The generation statement descends because, on the one hand, the sections over the intersection at a stage are finitely generated over the sections of one member, and on the other hand the elements of the limit which come from a stage inside the subring generated by the two members form a subring, which by hypothesis is everything. □

Theorem 11.5 (Separated descent of morphisms of finite presentation)**.** *Let (D_i) be a cofiltered inverse system of quasi-compact quasi-separated schemes with affine transition morphisms and let $g\colon X \to \varprojlim_i D_i$ be separated, quasi-compact and locally of finite presentation. Then there are an index i, a scheme Y and a morphism $p\colon Y \to D_i$ which is separated, quasi-compact and locally of finite presentation, together with a cartesian square exhibiting g as the base change of p along $\varprojlim_j D_j \to D_i$.*

Proof. Descend X by Theorem 11.3, then apply Theorem 11.4 to the diagonal of the descended model: a morphism of finite presentation whose base change to the limit is a closed immersion is a closed immersion at a finite stage. □

11.3 From finite presentation to finite type

Theorem 11.6 (Finite presentation approximation)**.** *Let S be quasi-compact quasi-separated and let $f\colon X \to S$ be locally of finite type with X quasi-compact quasi-separated. Then X is a cofiltered limit of schemes of finite presentation over S whose transition morphisms, and whose morphisms from X, are closed immersions.*

Sketch. Shrink an immersion of X into a finite-presentation model to a closed immersion $X \to Y$ into a quasi-compact open of that model, then write X as the limit of the closed subschemes of Y cut out by the finite-type subsheaves of the kernel of $X \to Y$. □

Theorem 11.7 (Separated finite presentation envelope)**.** *Let X and S be quasi-compact quasi-separated and let $f\colon X \to S$ be separated, quasi-compact and locally of finite type. Then f factors as*

$$X \xrightarrow{i} X' \xrightarrow{f'} S$$

with i a closed immersion and f' separated, quasi-compact and locally of finite presentation.

Proof. Take a finite presentation approximation of f (Theorem 11.6) and descend the separatedness of f to a stage of the system (Theorem 11.4); that stage is the required envelope. □

11.4 The case of a morphism of finite presentation

Theorem 11.8 (Nagata compactification for morphisms of finite presentation)**.** *Let S be quasi-compact quasi-separated and let $f\colon X \to S$ be separated, quasi-compact and locally of finite presentation. Then f admits a compactification.*

Proof. Write $S = \varprojlim_i D_i$ as a cofiltered limit of Noetherian stages with affine transition morphisms, by Theorem 11.2. By Theorem 11.5 the morphism f descends to a morphism $p\colon Y \to D_i$ over one of the stages which is again separated, quasi-compact and locally of finite presentation, with f the base change of p. The stage D_i is Noetherian, so p admits a compactification by Theorem 10.4. Base changing that compactification back along $S \to D_i$ compactifies f, by Theorem 2.9. □

12 The Nagata compactification theorem

Target Theorem 12.1 (Nagata compactification)**.** *Let Y be a quasi-compact quasi-separated scheme and let $f\colon X \to Y$ be separated, quasi-compact and locally of finite type. Then f factors as an open immersion followed by a proper morphism; that is, f admits a compactification in the sense of Definition 2.1.*

Proof. Since f is quasi-compact with quasi-compact quasi-separated target, the source X is quasi-compact quasi-separated as well. The morphism f is only locally of finite *type*, whereas Theorem 11.8 requires finite *presentation*; the bridge is the envelope of Theorem 11.7, which writes $f = f' \circ i$ with $i\colon X \to X'$ a closed immersion and $f'\colon X' \to Y$ separated, quasi-compact and locally of finite presentation. By Theorem 11.8 the morphism f' admits a compactification, and cutting that compactification back along the closed immersion i (Corollary 2.6) compactifies f. □

Theorem 12.1 is the goal of this lecture, and the surveyed material establishes it: every step of the chain

$$3.5 \;\Rightarrow\; 3.7, \qquad 7.7 \;\Rightarrow\; 8.1 \;\Rightarrow\; 9.12 \;\Rightarrow\; 10.4 \;\Rightarrow\; 11.8 \;\Rightarrow\; 12.1$$

has a complete proof, and no step is left as a hypothesis. What the argument nevertheless takes from outside is delimited in the next section.

13 What remains for the Nagata compactification theorem

13.1 No open steps

The chain leading to Target Theorem 12.1 contains no gap: there are *no* numbered open steps $G1, G2, \ldots$ in this lecture. Every statement above marked *Available* in the summary table of §14 has a complete proof, and Target Theorem 12.1 is deduced from those statements alone.

Two points deserve emphasis, since the classical treatment of the subject proceeds differently.

(1) *Elimination of indeterminacy is unconditional here.* The classical derivation of Theorem 8.1 rests on module flattening by blowing up, and thence on a dévissage which is not developed in the surveyed material. The route taken instead — through Theorem 7.7 and Zariski's Main Theorem, prepared by Theorems 7.1, 7.3, 7.4, 7.5 and 7.6 — avoids flattening altogether. Both external inputs of that route are available: Zariski's Main Theorem, and the immersion of a quasi-affine scheme into a relative affine space, which is Theorem 3.5 and is proved above.

(2) *The two-open theorem does not use flattening either.* It proceeds by boundary separation (Theorem 5.13), common modifications (Theorem 8.5) and the dense-source valuative criterion.

The surveyed material accordingly *does not* include a module-flattening theorem, nor the fibre-dimension package that usually accompanies it. These are not gaps in the proof of Target Theorem 12.1: they are simply absent, because the route taken does not require them.

13.2 Mathematical inputs taken from outside

The following results are used by the argument but are not established in the surveyed material. Each is stated in full at the place where it is first used; they are collected here only for reference, and they are the precise sense in which the proof above is not self-contained.

E1. Scheme-theoretic images.
Existence of the scheme-theoretic image of a quasi-compact morphism, its universal factorization property, and the fact that a quasi-compact immersion factors as an open immersion onto its scheme-theoretic image followed by a closed immersion. First used in Construction 2.4.

E2. Blow-ups.
Existence of the blow-up of a quasi-coherent ideal sheaf; its properness for a finite-type centre; principalization, i.e. invertibility of the inverse image of the centre on the blow-up; and the fact that the blow-up is an isomorphism over the complement of the centre. First used in §5.

E3. Zariski's Main Theorem.
In the two forms recalled before Theorem 7.7. Used exactly once, to pass from Theorem 7.6 to Theorem 7.7.

E4. Dense compactifications and boundary separation.
The replacement of a compactification by a scheme-theoretically dense one, and the separation of two disjoint constructible closed subsets of a quasi-compact open by an admissible blow-up. First used in Theorem 9.4.

E5. The dense-source valuative criterion.
Unique lifting of valuative squares whose generic point factors through a quasi-compact dominant morphism implies universal closedness and separatedness; together with the existence of valuation probes centred at points of a scheme-theoretic image. First used in Theorem 9.12.

13.3 Scope of the statements

Finally we record precisely how far each of the three principal compactification theorems reaches, since their hypotheses differ and the differences matter.

- Theorem 10.4 needs the *base* to be Noetherian; the morphism is separated, locally of finite type and quasi-compact.
- Theorem 11.8 needs the base only quasi-compact quasi-separated, but requires the morphism to be locally of finite *presentation*.
- Target Theorem 12.1 needs the base only quasi-compact quasi-separated and the morphism only locally of finite *type*, in addition to separated and quasi-compact.

Similarly, Theorem 7.7 assumes the preserved open U to be dense in X, whereas Theorem 7.5 — the case of a target that is quasi-affine over the base — does not; and Theorem 6.5 requires no finiteness hypothesis at all, whereas Theorems 5.5, 5.6 and 5.8 require the ambient scheme to be quasi-compact quasi-separated and the centres to be of finite type.

14 Main results

In the table below, *Available* means that the statement is proved in this lecture from earlier statements and from the inputs E1–E5 of §13; *External input* marks those five inputs themselves; and *Ultimate target* marks the goal of the lecture.

Node	Mathematical conclusion	Locator and status
Compactification	A factorization of a morphism as an open immersion followed by a proper morphism.	Definition 2.1; Available
Basic properties	A compactifiable morphism is separated and locally of finite type; the open immersion is quasi-compact iff the morphism is; for a proper morphism it is a closed immersion with clopen range.	Proposition 2.2; Available

Node	Mathematical conclusion	Locator and status
Immersion closure	A quasi-compact morphism factoring as an immersion into a scheme proper over the base is compactifiable.	Theorem 2.5; Available
Closed subschemes	Compactifiability of a quasi-compact morphism passes to closed subschemes of its source.	Corollary 2.6; Available
Base change	Compactifiability is stable under base change.	Theorem 2.9; Available
Projective space	$\mathbb{P}(\sigma; S) \to S$ is proper for finite σ; the standard chart $\mathbb{A}(\sigma; S) \to \mathbb{P}(\sigma; S)$ is an open immersion over S.	Theorem 3.2, Lemma 3.4; Available
Affine embedding	A quasi-affine scheme locally of finite type over S admits an immersion into a relative affine space over S.	Theorem 3.5; Available
Quasi-affine case	A quasi-compact, locally finite-type morphism with quasi-affine source is compactifiable.	Theorem 3.7; Available
Ideal extension	On a quasi-compact scheme, a finite-type ideal sheaf on a quasi-compact open extends to a finite-type ideal sheaf, inside a prescribed ambient one.	Theorem 4.3; Available
Boundary ideal	On a quasi-compact quasi-separated scheme, a quasi-compact open is the complement of the support of a finite-type ideal sheaf.	Theorem 4.6; Available
Compactness of ideals	A finite-type ideal sheaf is a compact element; every quasi-coherent ideal sheaf is the directed supremum of its finite-type subsheaves.	Theorem 4.5; Available
Power in an ideal	$V(\mathcal{A}) \subseteq V(\mathcal{B})$ with $\mathcal{B}$ of finite type implies $\mathcal{B}^n \leq \mathcal{A}$ for some n.	Lemma 4.7; Available
Admissible centre	A finite-type centre whose support misses a prescribed open; the blow-up is proper and an isomorphism over that open.	Definition 5.1, Proposition 5.2; Available
Universal property	A morphism whose inverse image of the centre is invertible factors uniquely through the blow-up.	Theorem 5.3; Available
Product centres	$\mathrm{Bl}_{\mathcal{I}\mathcal{L}}(X) \cong \mathrm{Bl}_{\mathcal{L}\cdot\mathcal{O}}(\mathrm{Bl}_{\mathcal{I}} X)$; invertible factors do not change a blow-up.	Theorem 5.4; Available
Finite degree	A finite-type centre on a blow-up is captured by a finite-type $\mathcal{L} \leq \mathcal{I}^d$ downstairs.	Theorem 5.5; Available
Composition	A blow-up of a blow-up is a single blow-up, with support the union of the two supports; admissibility is preserved.	Theorem 5.6, Corollary 5.7; Available
Centre extension	A finite-type centre on a quasi-compact open extends to an admissible centre whose blow-up restricts to the given one.	Theorem 5.8; Available
Density transport	Admissible blow-ups preserve density and scheme-theoretic density; one admissible blow-up makes the preserved open schematically dense.	Theorems 5.10, 5.11; Available
Boundary separation	Strict transforms of $V(\mathcal{I}_1)$ and $V(\mathcal{I}_2)$ along the blow-up of $\mathcal{I}_1 \sqcup \mathcal{I}_2$ are disjoint.	Theorem 5.13; Available
Cartier boundary	After an admissible blow-up the boundary supports an invertible ideal sheaf.	Theorem 5.14; Available
Strict transform	The scheme-theoretic image of the locus over the preserved open, with its inherited properties and range bounds.	Definition 6.1, Proposition 6.2; Available
Strict transform is a blow-up	The strict transform of f along $\mathrm{Bl}_{\mathcal{I}}(X)$ is $\mathrm{Bl}_{\mathcal{I}\cdot\mathcal{O}_Y}(Y)$; no finiteness hypotheses.	Theorem 6.5, Corollary 6.6; Available
Disjointness blow-up	After blowing up $\mathcal{I}_1 \sqcup \mathcal{I}_2$ the image of Y misses the part over $\overline{V \setminus U}$.	Theorem 7.1; Available
Conrad, Case 1	Extension of a partial section into a relative affine space, and into a quasi-affine target, over an affine base.	Theorems 7.2, 7.3; Available
Conrad, Case 2	The same over a quasi-compact quasi-separated base for a quasi-affine morphism, with $U = V$.	Theorem 7.4; Available

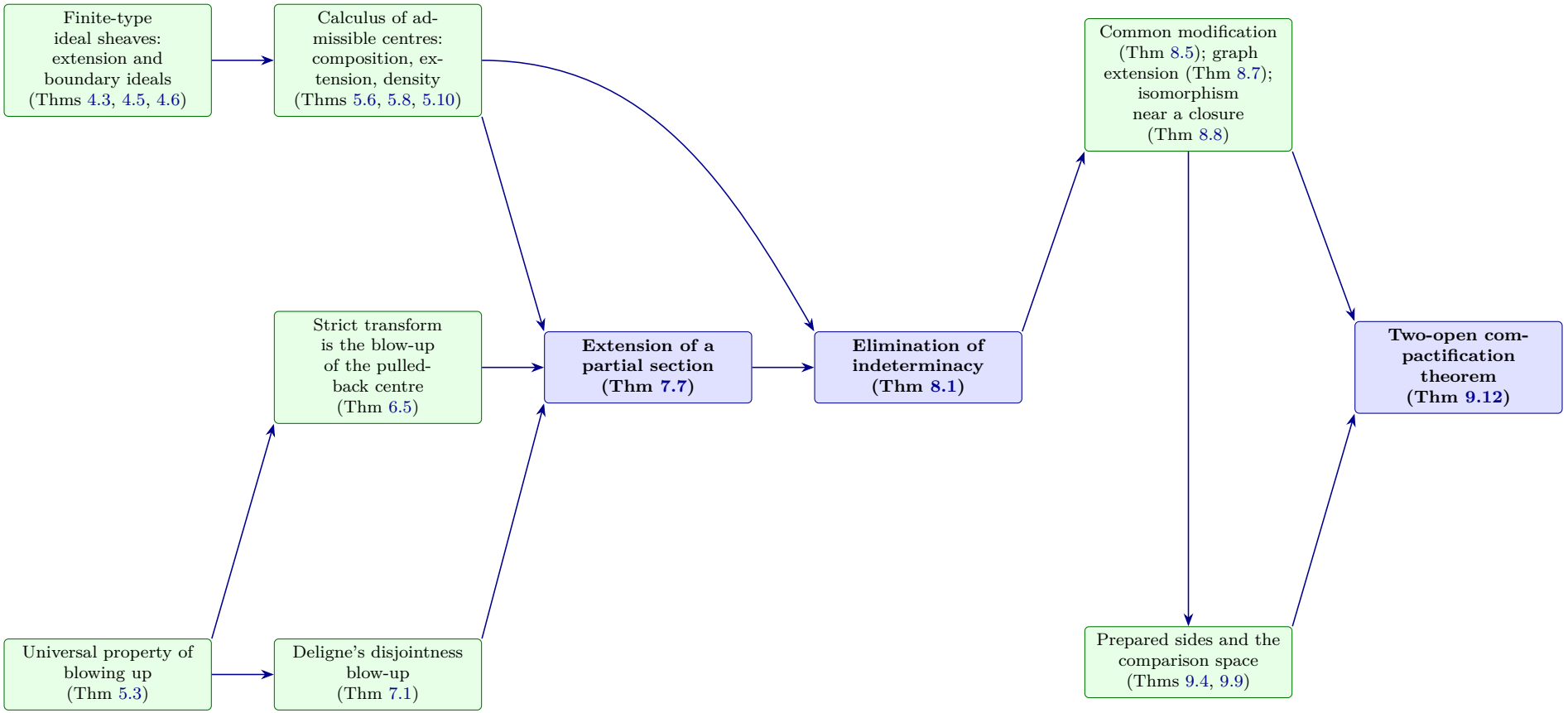
Node	Mathematical conclusion	Locator and status
Conrad, Case 3	The same for two nested quasi-compact opens $U \leq V$; density of U not assumed.	Theorem 7.5; Available
Quasi-finite graph	After one admissible blow-up the scheme-theoretic image of the graph is quasi-finite and separated over the blow-up.	Theorem 7.6; Available
Partial section extension	A partial section over a dense quasi-compact open, with closed image, extends after an admissible blow-up, with closed graph.	Theorem 7.7, Corollary 7.8; Available
Elimination of indeterminacy	After an admissible blow-up, the strict transform of a separated finite-type morphism which is an isomorphism over U is an open immersion.	Theorem 8.1; Available
Image transport	Scheme-theoretic images are carried to strict transforms by admissible blow-ups.	Theorem 8.3; Available
Common modification	Two separated finite-type models of a common open admit admissible blow-ups gluing to a separated finite-type model.	Definition 8.4, Theorem 8.5; Available
Graph extension	A compactification of a quasi-compact open $V \subseteq U$ extends to a proper morphism $V^{\mathrm{gr}} \to U$ restricting to the identity on V.	Definition 8.6, Theorem 8.7; Available
Isomorphism near a closure	A proper morphism which is an isomorphism near a closed piece of U acquires a strict transform which is an isomorphism near its closure.	Theorem 8.8; Available
Prepared sides	Proper ambient models with disjoint self and forbidden boundaries exist for each compactifiable open.	Definitions 9.1, 9.2, Theorem 9.4; Available
Comparison space	The scheme-theoretic image of $V_1 \times_U V_2$ in $X_1 \times_S X_2$, with proper projections; key boundary disjointness.	Definitions 9.5, 9.6, Theorems 9.8, 9.9; Available
Intermediate schemes	$W = U \sqcup_O X^\circ$ is separated over S and contains a dense copy of U.	Definition 9.10, Proposition 9.11; Available
Two-open theorem	A separated finite-type morphism to a Noetherian base is compactifiable if its source is covered by two compactifiable opens with dense intersection.	Theorem 9.12; Available
Dense affine cover	A quasi-compact scheme with finitely many irreducible components has a finite affine open cover with dense total intersection.	Theorem 10.1; Available
Noetherian case	A separated, quasi-compact morphism locally of finite type to a Noetherian scheme is compactifiable.	Theorem 10.4; Available
Noetherian approximation	Every quasi-compact quasi-separated scheme is a cofiltered limit of finite-type $\mathbb{Z}$-schemes with affine transition morphisms.	Definition 11.1, Theorem 11.2; Available
Finite-presentation descent	A separated morphism of finite presentation over such a limit descends to a stage, with the same properties.	Theorems 11.3, 11.4, 11.5; Available
Finite-type envelope	A separated, quasi-compact, locally finite-type morphism of quasi-compact quasi-separated schemes is a closed immersion followed by a separated morphism of finite presentation.	Theorems 11.6, 11.7; Available
Finite presentation case	A separated, quasi-compact morphism of finite presentation to a quasi-compact quasi-separated base is compactifiable.	Theorem 11.8; Available
E1 Scheme images	Existence and universal property of scheme-theoretic images of quasi-compact morphisms.	§13, used from Construction 2.4; External input
E2 Blow-ups	Existence, properness, principalization, isomorphism over the complement of the centre.	§13, used from §5; External input

Node	Mathematical conclusion	Locator and status
E3 Zariski's Main Theorem	A separated, quasi-compact, locally quasi-finite morphism of finite type is quasi-affine as a morphism.	§13, used in Theorem 7.7; External input
E4 Dense refinement and boundary separation	Dense refinement of a compactification; separation of disjoint constructible closed subsets by an admissible blow-up.	§13, used in Theorem 9.4; External input
E5 Dense-source valuative criterion	Unique valuative lifting from a dense source gives universal closedness and separatedness.	§13, used in Theorem 9.12; External input
Nagata compactification	A separated, quasi-compact morphism locally of finite type over a quasi-compact quasi-separated base factors as an open immersion followed by a proper morphism.	Target Theorem 12.1; **Ultimate target**

15 Dependency mind map

The two diagrams below display the same dependency relation $A \to B$ = "the conclusion B uses the conclusion A". Panel A shows the dependencies internal to the lecture that lead to the two-open compactification theorem; Panel B shows the route from the available ingredients to the target theorem, together with the five inputs taken from outside.

Panel A. Established dependencies



Legend. established conclusion · decisive output of a branch · → "A is used by B"

Panel B. From the available ingredients to the target

